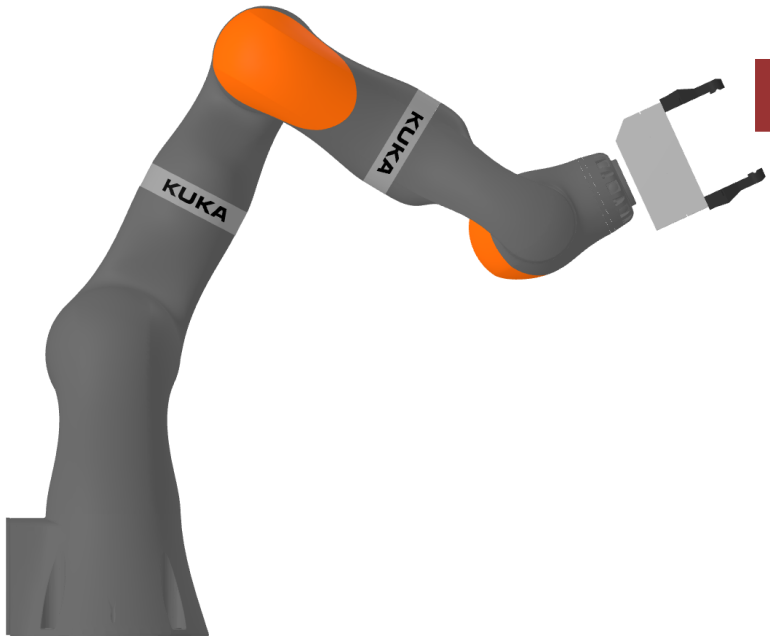


Manipulator Control

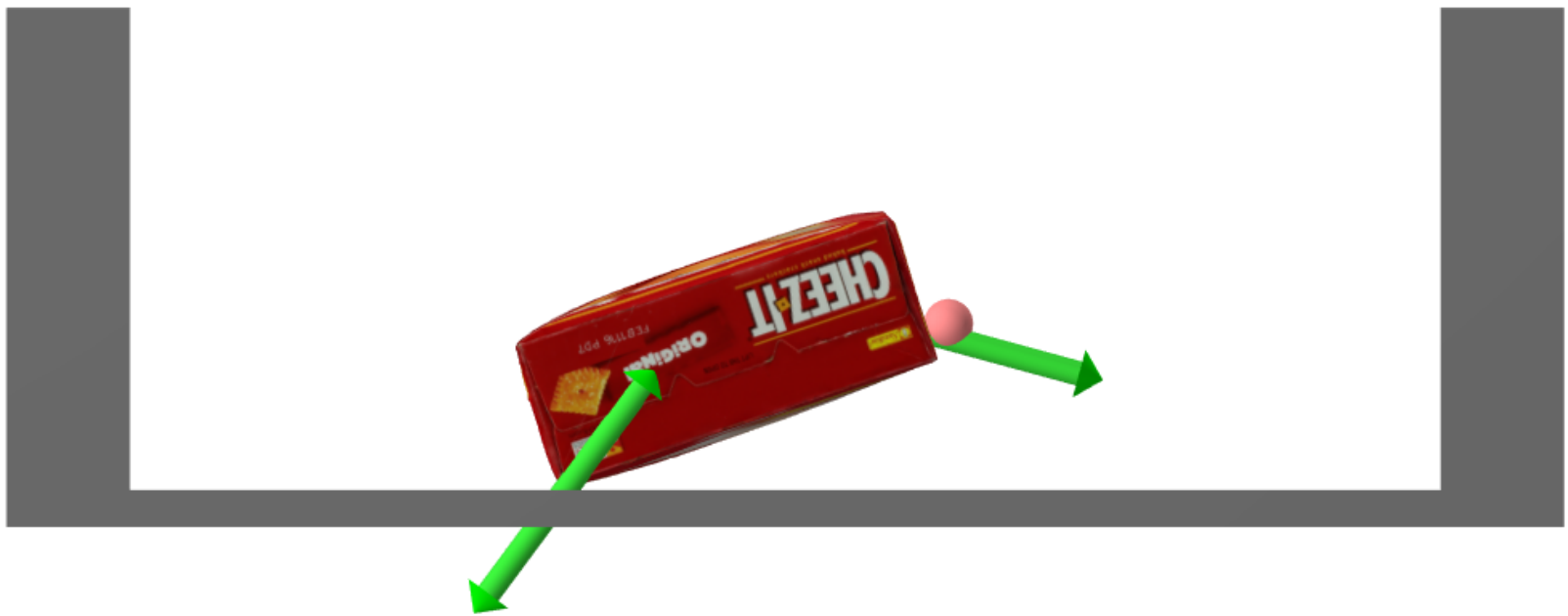
MIT 6.4210/2:

Robotic Manipulation

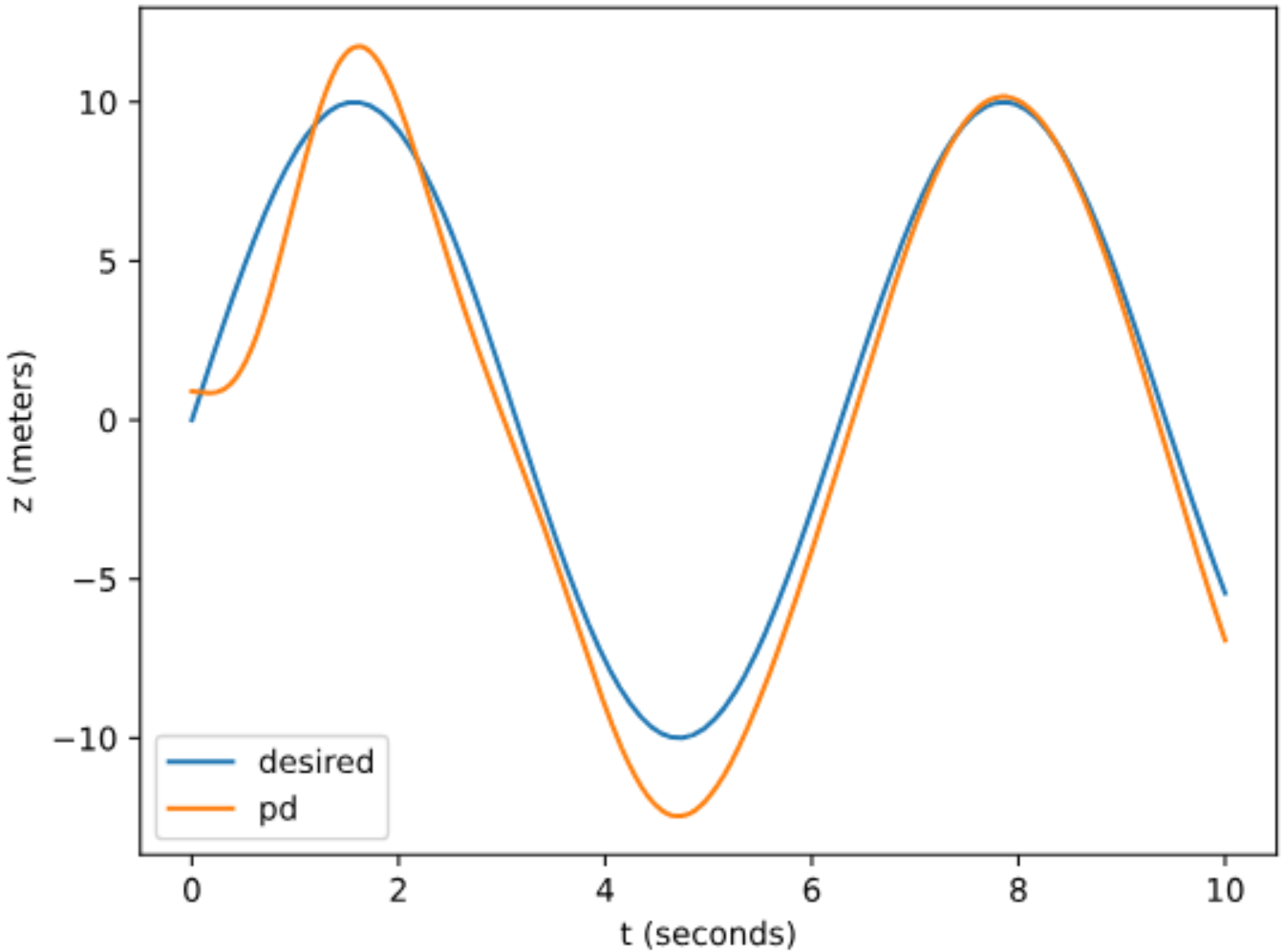
Fall 2025, Lecture 14

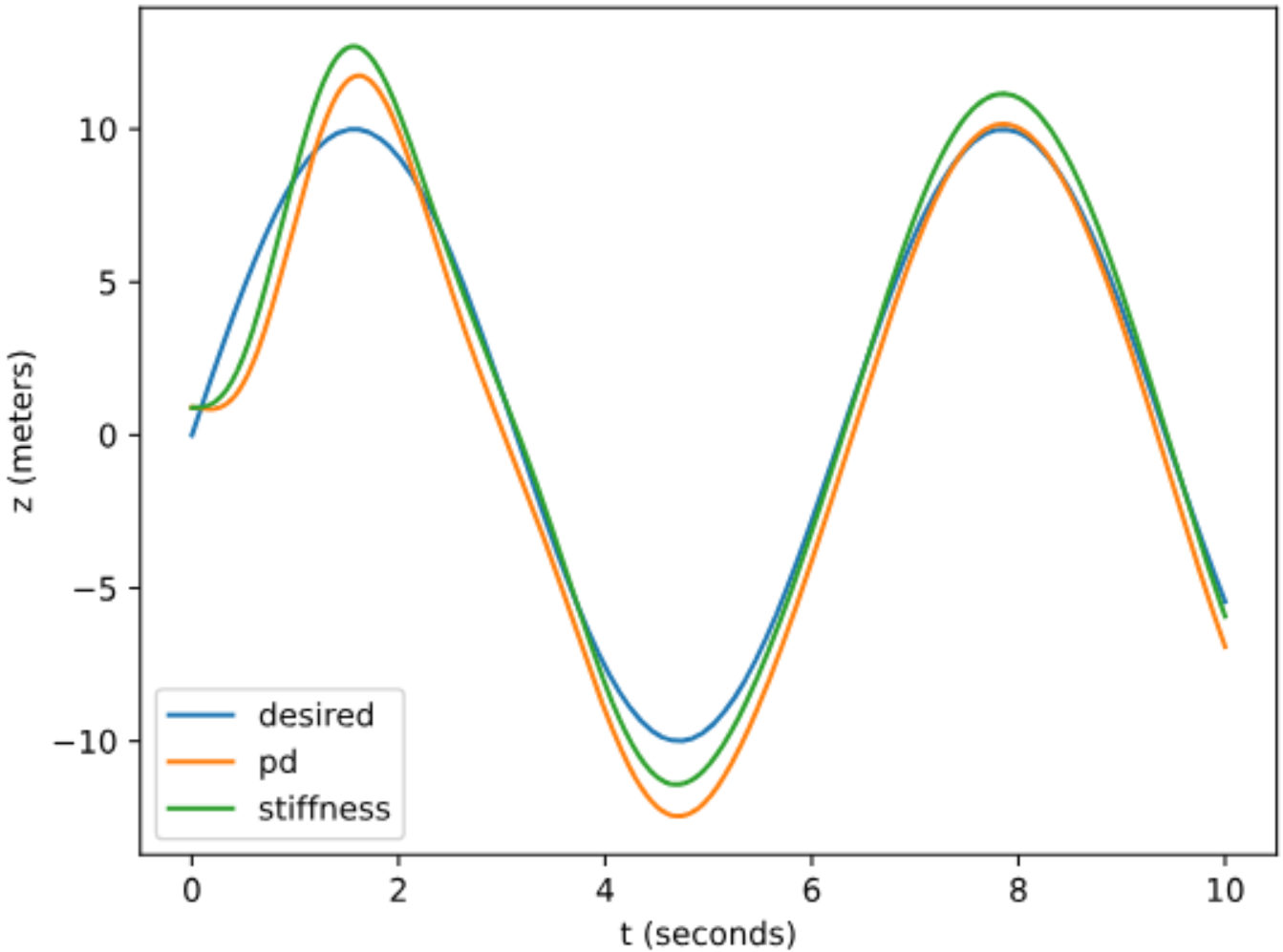
Massachusetts Institute of Technology

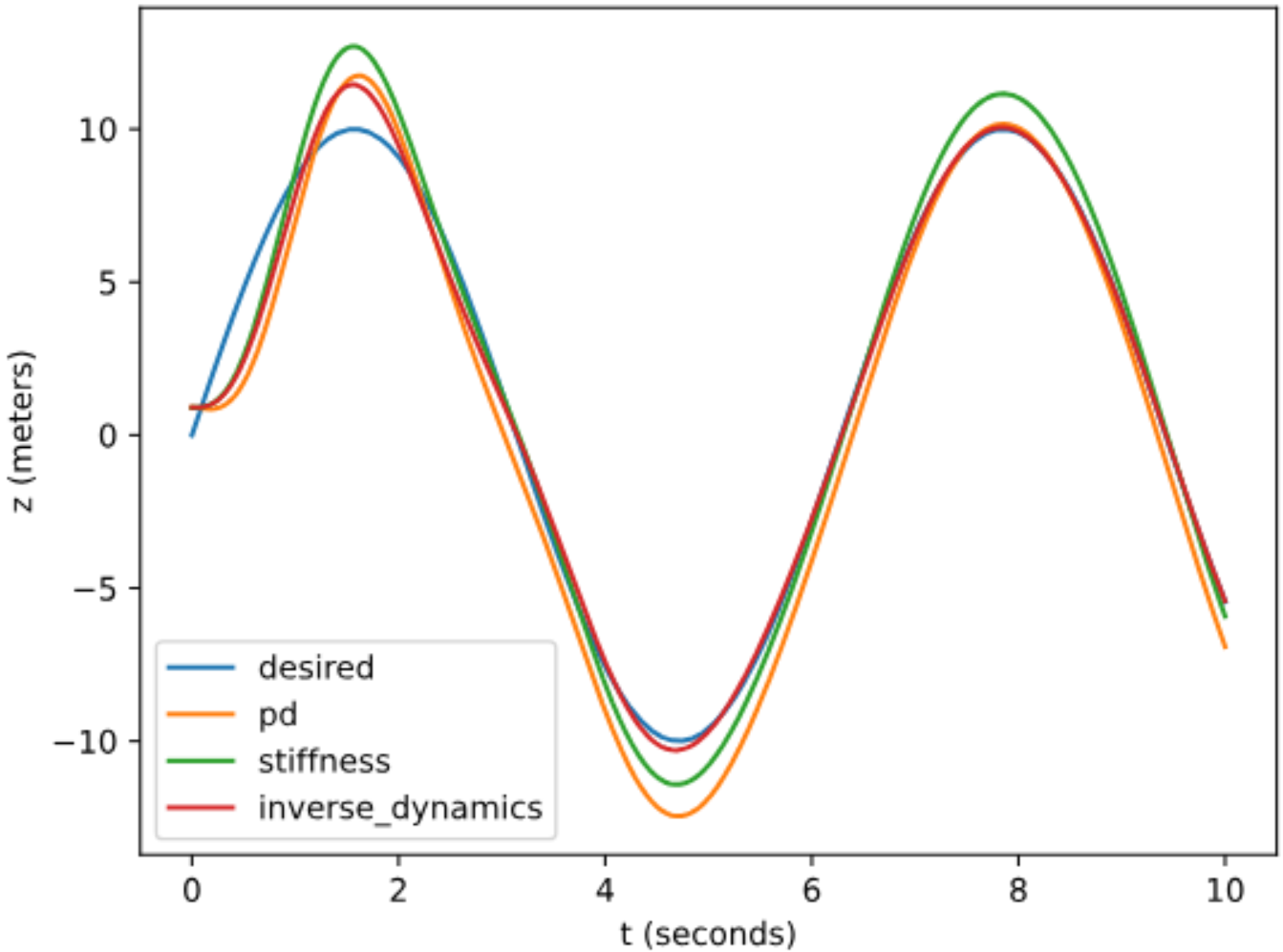
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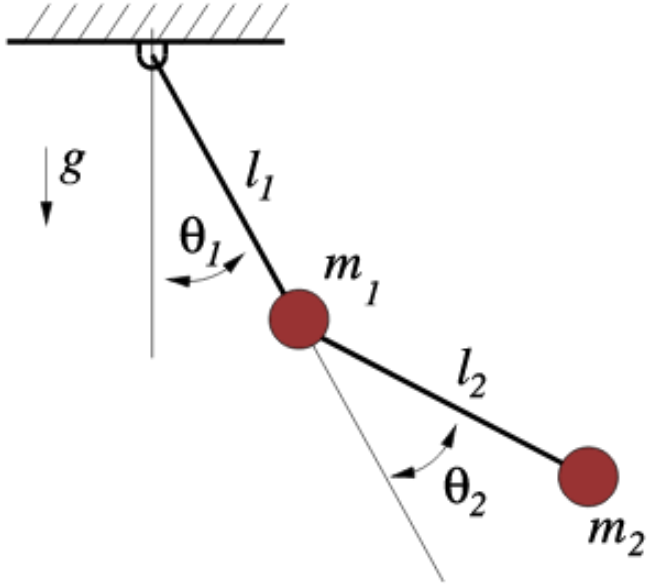
https://manipulation.csail.mit.edu/force.html#systems


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desired
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stiffness
inverse_dynamics

https://manipulation.csail.mit.edu/force.html#systems



Full derivation is here.

$$M(q) = \begin{bmatrix} (m_1+m_2)l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 c_2 & m_2 l_2^2 + m_2 l_1 l_2 c_2 \\ m_2 l_2^2 + m_2 l_1 l_2 c_2 & m_2 l_2^2 \end{bmatrix}$$

$$C(q,\dot{q}) = \begin{bmatrix} 0 & -m_2 l_1 l_2 (2\dot{q}_1 + \dot{q}_2) s_2 \\ m_2 l_1 l_2 \dot{q}_1 s_2 & 0 \end{bmatrix}$$

$$\tau_g(q) = -g \begin{bmatrix} (m_1+m_2) l_1 s_1 + m_2 l_2 s_{1+2} \\ m_2 l_2 s_{1+2} \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



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https://manipulation.csail.mit.edu/force.html#systems

https://www.youtube.com/embed/lZvPslCIQZY?mute=1&rel=0&start=6&end=57&enablejsapi=1

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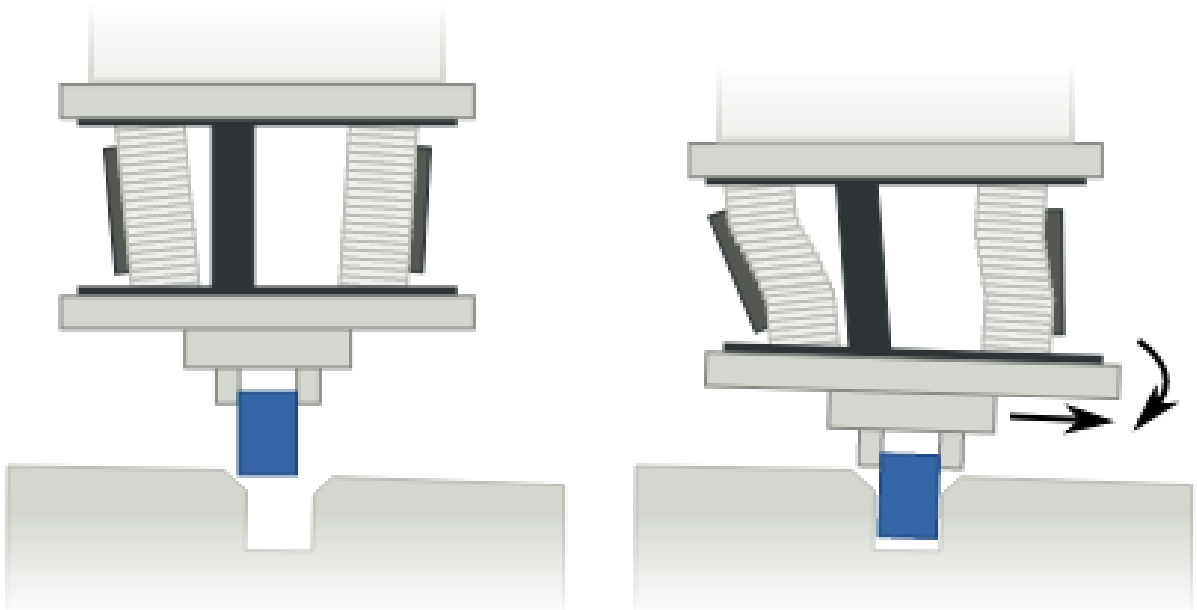


Image credit for Arne Nordmann, 2008